Lab 2

Four to Sixteen Decoder

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Group U

ECE-3300L
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Design:

Gate Implementation:

```
`timescale 1ns / 1ps

module decoder4x16_gate(
    input wire [3:0] A, input wire E,
    output wire [15:0] Y
    );

    wire a0 = A[0]; wire a0n = ~a0;
    wire a1 = A[1]; wire a1n = ~a1;
    wire a2 = A[2]; wire a2n = ~a2;
    wire a3 = A[3]; wire a3n = ~a3;

    assign Y[0] = a0n & a1n & a2n & a3n & E;
    assign Y[1] = a0 & a1n & a2n & a3n & E;
    assign Y[2] = a0n & a1 & a2n & a3n & E;
    assign Y[3] = a0 & a1 & a2n & a3n & E;
    assign Y[4] = a0n & a1n & a2 & a3n & E;
    assign Y[5] = a0 & a1n & a2 & a3n & E;
    assign Y[6] = a0n & a1 & a2 & a3n & E;
    assign Y[7] = a0 & a1 & a2 & a3n & E;
    assign Y[8] = a0n & a1n & a2n & a3 & E;
    assign Y[9] = a0 & a1n & a2n & a3 & E;
    assign Y[10] = a0n & a1 & a2n & a3 & E;
    assign Y[11] = a0 & a1 & a2n & a3 & E;
    assign Y[12] = a0n & a1n & a2 & a3 & E;
    assign Y[13] = a0 & a1n & a2 & a3 & E;
    assign Y[14] = a0n & a1 & a2 & a3 & E;
    assign Y[15] = a0 & a1 & a2 & a3 & E;

endmodule
```

Behavioural Implementation:

```
module decoder4x16_behav(
    input wire [3:0] A, input wire E,
    output wire [15:0] Y
        );

   reg [15:0]destin;
   integer i;
```

```
always @(*) begin
    destin = 16'b0;                              //reset
    if(E)begin                                   //enable check
        for(i = 0; i < 16; i = i + 1)begin
            destin[i] = (A == i);               //case mapping
        end
    end
end
assign Y = destin;
endmodule
```

XDC Snippet:

```
##Switches

set_property -dict { PACKAGE_PIN J15   IOSTANDARD LVCMOS33 } [get_ports { A[0] }];      // SW0
set_property -dict { PACKAGE_PIN L16   IOSTANDARD LVCMOS33 } [get_ports { A[1] }];      // SW1
set_property -dict { PACKAGE_PIN M13   IOSTANDARD LVCMOS33 } [get_ports { A[2] }];      // SW2
set_property -dict { PACKAGE_PIN R15   IOSTANDARD LVCMOS33 } [get_ports { A[3] }];      // SW3
set_property -dict { PACKAGE_PIN R17   IOSTANDARD LVCMOS33 } [get_ports { E }];         // SW4

## LEDs

set_property -dict { PACKAGE_PIN H17   IOSTANDARD LVCMOS33 } [get_ports { Y[0] }];      // LED0
set_property -dict { PACKAGE_PIN K15   IOSTANDARD LVCMOS33 } [get_ports { Y[1] }];      // LED1
set_property -dict { PACKAGE_PIN J13   IOSTANDARD LVCMOS33 } [get_ports { Y[2] }];      // LED2
…                                                                                        …
set_property -dict { PACKAGE_PIN V14   IOSTANDARD LVCMOS33 } [get_ports { Y[13] }];     // LED13
set_property -dict { PACKAGE_PIN V12   IOSTANDARD LVCMOS33 } [get_ports { Y[14] }];     // LED14
set_property -dict { PACKAGE_PIN V11   IOSTANDARD LVCMOS33 } [get_ports { Y[15] }];     // LED15
```

Simulation:

Description:

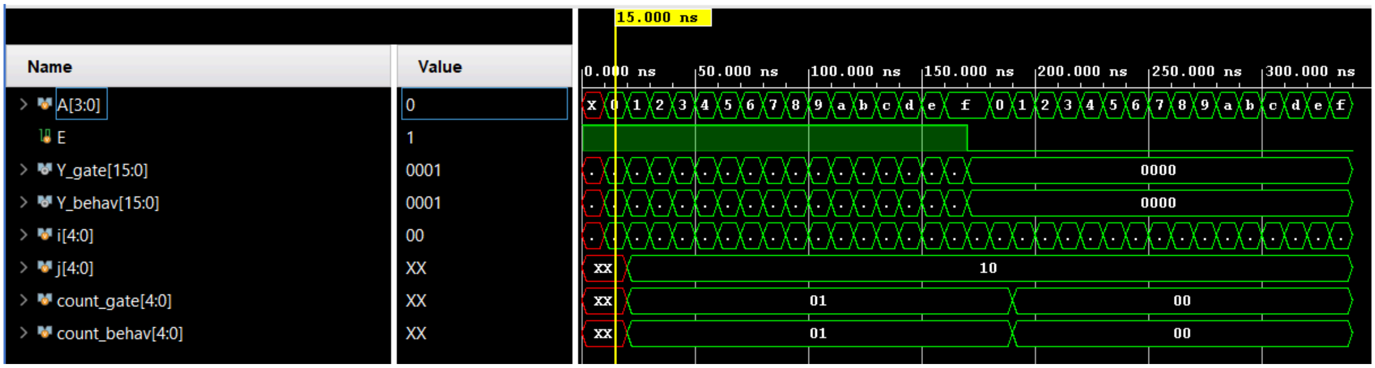
For the testbench I used a nested for loop to automatically verify only one Y[i] == 1. The outside loop iterates through each possible value of the input (A is 4 bits wide so values 0 to 15), and the inside loop iterates through each individual bit of the output (Y[j], 16 total), keeping a counter to track when Y[j] == 1.

Both implementations are tested simultaneously with both cases of enable. First, with the decoder enabled, the test fails automatically when the counter does not equal one, or the output bit does not match the input value. Then, with the decoder disabled, the test fails if the counter does not equal zero. Both tests display a pass or fail message, with a final message when all tests pass.

Output:

```
# run 1000ns
Decoder enabled
PASS: E=1 A=0000 Y_gate=0000000000000001 Y_behav=0000000000000001
PASS: E=1 A=0001 Y_gate=0000000000000010 Y_behav=0000000000000010
PASS: E=1 A=0010 Y_gate=0000000000000100 Y_behav=0000000000000100
PASS: E=1 A=0011 Y_gate=0000000000001000 Y_behav=0000000000001000
PASS: E=1 A=0100 Y_gate=0000000000010000 Y_behav=0000000000010000
PASS: E=1 A=0101 Y_gate=0000000000100000 Y_behav=0000000000100000
PASS: E=1 A=0110 Y_gate=0000000001000000 Y_behav=0000000001000000
PASS: E=1 A=0111 Y_gate=0000000010000000 Y_behav=0000000010000000
PASS: E=1 A=1000 Y_gate=0000000100000000 Y_behav=0000000100000000
PASS: E=1 A=1001 Y_gate=0000001000000000 Y_behav=0000001000000000
PASS: E=1 A=1010 Y_gate=0000010000000000 Y_behav=0000010000000000
PASS: E=1 A=1011 Y_gate=0000100000000000 Y_behav=0000100000000000
PASS: E=1 A=1100 Y_gate=0001000000000000 Y_behav=0001000000000000
PASS: E=1 A=1101 Y_gate=0010000000000000 Y_behav=0010000000000000
PASS: E=1 A=1110 Y_gate=0100000000000000 Y_behav=0100000000000000
PASS: E=1 A=1111 Y_gate=1000000000000000 Y_behav=1000000000000000
Decoder disabled
PASS: E=0 A=0000 Y_gate=0000000000000000 Y_behav=0000000000000000
PASS: E=0 A=0001 Y_gate=0000000000000000 Y_behav=0000000000000000
PASS: E=0 A=0010 Y_gate=0000000000000000 Y_behav=0000000000000000
PASS: E=0 A=0011 Y_gate=0000000000000000 Y_behav=0000000000000000
PASS: E=0 A=0100 Y_gate=0000000000000000 Y_behav=0000000000000000
PASS: E=0 A=0101 Y_gate=0000000000000000 Y_behav=0000000000000000
PASS: E=0 A=0110 Y_gate=0000000000000000 Y_behav=0000000000000000
PASS: E=0 A=0111 Y_gate=0000000000000000 Y_behav=0000000000000000
PASS: E=0 A=1000 Y_gate=0000000000000000 Y_behav=0000000000000000
PASS: E=0 A=1001 Y_gate=0000000000000000 Y_behav=0000000000000000
PASS: E=0 A=1010 Y_gate=0000000000000000 Y_behav=0000000000000000
PASS: E=0 A=1011 Y_gate=0000000000000000 Y_behav=0000000000000000
PASS: E=0 A=1100 Y_gate=0000000000000000 Y_behav=0000000000000000
PASS: E=0 A=1101 Y_gate=0000000000000000 Y_behav=0000000000000000
PASS: E=0 A=1110 Y_gate=0000000000000000 Y_behav=0000000000000000
PASS: E=0 A=1111 Y_gate=0000000000000000 Y_behav=0000000000000000
All Tests Passed.
$finish called at time : 340 ns : File "D:/School/ECE3300/ECE3300_Lab2_GroupU
```

Waveform:



Implementation:

Name	Slice LUTs (63400)	Slice (15850)	LUT as Logic (63400)	Bonded IOB (210)
N decoder4x16_gate	8	4	8	21

Design Timing Summary

Setup		**Hold**		**Pulse Width**	
Worst Negative Slack (WNS):	inf	Worst Hold Slack (WHS):	inf	Worst Pulse Width Slack (WPWS):	NA
Total Negative Slack (TNS):	0.000 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	NA
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	NA
Total Number of Endpoints:	16	Total Number of Endpoints:	16	Total Number of Endpoints:	NA

There are no user specified timing constraints.

Name	Slice LUTs (63400)	Slice (15850)	LUT as Logic (63400)	Bonded IOB (210)
N decoder4x16_behav	8	4	8	21

Design Timing Summary

Setup		**Hold**		**Pulse Width**	
Worst Negative Slack (WNS):	inf	Worst Hold Slack (WHS):	inf	Worst Pulse Width Slack (WPWS):	NA
Total Negative Slack (TNS):	0.000 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	NA
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	NA
Total Number of Endpoints:	16	Total Number of Endpoints:	16	Total Number of Endpoints:	NA

There are no user specified timing constraints.

Group Video:

https://www.youtube.com/watch?v=XRHvgI2aRyE

Contributions:

Ben Robles: XDC file, simulation
Robert Stevenson: Design implemenation

Reflections:

Ben Robles: This lab was a good first step into both aspects of digital circuit design. It was interesting to see the different methods of implementation and work with them first hand to discover their differences. For the simulation portion, I enjoyed the less hand-holding, create it yourself methodology here and it helped me better understand how to write an in depth testbench.

Robert Stevenson: This lab was gave me the opportunity to work in both behavioral and gate level implementations of a four to sixteen decoder, as well as further experience working in Vivado and debugging hardware design.